

WLM FOUNDATION SERIES
COMPLETE EDITION

Papers 0 through 5

Wujie Gu

World-Layer Model (WLM) Series • April 2026
OSF: osf.io/rq5vj • ResearchGate

Copyright © 2026 Wujie Gu. NCND License. Non-commercial sharing in original form only. No modifications or derivative works without explicit written consent.

Series Overview

The WLM Foundation Series establishes the minimal generative structure required for any universe to exist. It proves, through six successive papers, that the chain:

$0 \rightarrow \epsilon \rightarrow \pm\epsilon \rightarrow \text{Recursion} \rightarrow \text{Structure} \rightarrow \text{Appearance}$

is forced, unique, and unskippable at every step. No link can be removed. No link can be reordered. Each paper adds one confirmed unskippability result toward Conjecture U1 (the full chain uniqueness conjecture).

The series uses five axioms: S0 (Symmetry Closure), D0 (Generative Orientation and Reversal), R1 (Composability), R2 (Recursive Orientation Preservation), and P0 (Projection Existence), plus V-WF (Well-Founded Valuation). No other assumptions are made. All theorems follow from these axioms and prior results within the series.

Paper	Title	Forced Step	Unskippability Result
0	The Generative Beginning of a Universe	Full chain overview	Conjecture U1 (program)
1	The Stability and Uniqueness of 0	$0 \rightarrow \epsilon$	Thm 3.3: ϵ unskippable
2	The Uniqueness and Unskippability of $\pm\epsilon$	$\epsilon \rightarrow \{\pm\epsilon\}$	Thm 4.3: $\pm\epsilon$ unskippable
3	The Uniqueness and Unskippability of Recursion	$\{\pm\epsilon\} \rightarrow \text{Recursion}$	Thm 4.3: Recursion unskippable

4	The Uniqueness and Unskippability of Invariant Structure	Recursion $\rightarrow$ Structure	Thm 4.3: Structure unskippable
5	The Uniqueness and Unskippability of Appearance	Structure $\rightarrow$ Appearance	Thm 4.3: Appearance unskippable

Open problem. Conjecture U1 (Chain Uniqueness): under axioms $S0, D0, R1, R2, P0, V-WF$, the chain $0 \rightarrow \varepsilon \rightarrow \{\pm\varepsilon\} \rightarrow \text{recursion} \rightarrow \text{structure} \rightarrow \text{appearance}$ is the unique forced generative sequence with no alternative paths. The five unskippability results confirm that the chain cannot be shortened. The full proof that no alternative chain of the same length exists remains the central open problem of the series.

How to Read This Edition

Each paper is self-contained and begins with a section listing its inherited results from prior papers. The papers may be read sequentially (recommended for first reading) or individually (each paper states its dependencies explicitly).

The Complete Edition appends: (A) a Unified Glossary defining all terms used across the series; (B) an Axiom–Theorem Dependency Matrix showing which axioms each theorem depends on; (C) a Unified References section consolidating all citations. Individual papers do not repeat these appendices.

Notation is consistent throughout. The main symbols: X = state space; 0 = perfect symmetry state; ε = minimal offset; $\pm\varepsilon$ = dual pair; I_0 = symmetry interval; $\mathcal{G}$ = generative relations; $v(\cdot)$ = distinguishability valuation; $\sim_o$ = observational equivalence; $\theta(\cdot)$ = orientation label; S = invariant structure; P = projection; Σ = appearance layer.

Theorem numbering: each paper uses its own local numbering (e.g., Paper 2 Theorem 3.1). Cross-references always include the paper number (e.g., “Paper 1 Theorem 3.3”). The dependency matrix in the appendix shows all cross-paper dependencies.

WLM Foundation Series

Paper 0**The Generative Beginning of a Universe:** $0 \rightarrow \varepsilon \rightarrow \pm\varepsilon \rightarrow \text{Recursion} \rightarrow \text{Structure} \rightarrow \text{Appearance}$ **Abstract**

This paper establishes the minimal generative structure required for any universe to exist. It does not describe the Big Bang, inflation, or quantum gravity. It addresses a prior question: what structural conditions must hold for any world to exist at all? The chain $0 \rightarrow \varepsilon \rightarrow \pm\varepsilon \rightarrow \text{recursion} \rightarrow \text{structure} \rightarrow \text{appearance}$ is introduced, and each step is grounded in five explicit axioms. Papers 1–5 each establish one link of this chain with full proofs.

Epistemic status. Theorems are provable within the WLM framework given axioms S0, D0, R1, R2, P0. The chain uniqueness conjecture (Conjecture U1) is a research program. Partial results are confirmed in Papers 1–5.

 $0 \rightarrow \varepsilon \rightarrow \pm\varepsilon \rightarrow \text{recursion} \rightarrow \text{structure} \rightarrow \text{appearance}$ **1. Axiom System**

Axiom S0 (Symmetry Closure). If a state has no distinguishable relations to any other state, no generative rule can produce a distinguishable successor. Any perfectly indistinct state is a fixed point of every generative rule.

Axiom D0 (Generative Orientation and Reversal). (a) *Primitive Orientation:* any state $x \neq_0 0$ carries an orientation label $\theta(x)$, preserved under observational equivalence. (b) *Orientation Reversal:* there exists at least one generator F realizing orientation reversal on ε .

Axiom R1 (Composability). If $G, H \in \mathcal{G}$, then $H \circ G \in \mathcal{G}$. All iterates f^n are defined for all $n \geq 1$.

Axiom R2 (Recursive Orientation Preservation). *If f realizes a directional flip, all iterates f^n preserve the alternation pattern. No recursion can wash out the $\pm\epsilon$ directional structure.*

Axiom P0 (Projection Existence). *Any WLM system with stable invariants admits at least one observational layer Σ and a projection $P: D \rightarrow \Sigma$ preserving the structural relations of those invariants.*

Axiom V-WF (Well-Founded Valuation). *V is a well-founded partially ordered set with minimal element $v(0)$. Every nonempty subset of $\{v(x): v(x) > v(0)\}$ has a minimal element.*

2. The Generative Chain: Sketch

Each step in the chain is established in full in a dedicated paper. This section gives the structural logic of each transition.

Step 1: $0 \rightarrow \epsilon$ (Paper 1)

The state 0 has no distinguishable relations. By S0, it is a fixed point of every generative rule: no universe can begin at 0. The minimal departure from 0 is the minimal offset ϵ — the unique (up to observational equivalence) first distinguishable state. ϵ exists if and only if the system generates non-trivial structure, and no chain can skip it.

Step 2: $\epsilon \rightarrow \pm\epsilon$ (Paper 2)

Once ϵ exists, D0 forces orientation: ϵ carries a unique primitive orientation. The dual direction $-\epsilon$ must also exist. The orientation space at valuation $v(\epsilon)$ is binary: exactly the pair $\{+\epsilon, -\epsilon\}$. No third primitive direction is possible. Duality is unskippable.

Step 3: $\pm\epsilon \rightarrow \text{Recursion}$ (Paper 3)

Duality alone is not enough. A system that can only alternate between $+\epsilon$ and $-\epsilon$ finitely many times will stagnate. Persistent alternation requires infinite iteration — recursion. Under R1 and R2, recursion is the unique engine capable of sustaining $\pm\epsilon$ alternation indefinitely. Recursion is unskippable.

Step 4: $\text{Recursion} \rightarrow \text{Structure}$ (Paper 4)

Any non-collapsing recursive orbit generates at least one invariant structure: a periodic cycle or a compact closure (Bolzano-Weierstrass). This structure is unique at the minimal valuation level. Structure is unskippable for any chain leading to appearance.

Step 5: Structure → Appearance (Paper 5)

By P0, any stable invariant structure admits a projection to an observational layer. This projection is appearance — the phenomenal world-layer. Without structure, there is nothing to project. Appearance is the final and unskippable step.

3. Collapse Modes

At each step, failure to establish the required layer leads to collapse. The series classifies all collapse modes systematically:

- Symmetry collapse (Paper 1): no departure from 0; all states remain indistinguishable.
- Directionless collapse (Paper 2): ϵ exists but carries no orientation; no duality.
- Asymmetry collapse (Paper 2): only one direction realized; no alternation target.
- Static collapse (Paper 3): no recursion; system stagnates at $\pm\epsilon$.
- Divergent collapse (Paper 3): non-closed iteration; no invariants.
- Oscillation collapse (Paper 4): recursion without invariant closure.
- Pre-appearance collapse (Paper 5): invariants without projection.

Every chain that fails to produce a world-layer falls into one of these modes.

4. Comparison with Physical Cosmology

Physical cosmology describes the temporal evolution of a specific universe from an initial state. WLM describes the minimal structural conditions for any universe to exist at all. These are complementary accounts. WLM makes no claims about the Big Bang, inflation, quantum gravity, or specific physical constants. Those belong to the temporal account of a particular physical universe. WLM provides the structural ground for why any such universe is possible.

5. Conclusion

Paper 0 introduces the chain $0 \rightarrow \varepsilon \rightarrow \pm\varepsilon \rightarrow \text{recursion} \rightarrow \text{structure} \rightarrow \text{appearance}$ and the five axioms that ground it. The complete proofs are developed in Papers 1–5. The chain uniqueness conjecture (U1) is the central open problem: that this chain is not only forced but uniquely forced, with no alternative route of the same length. The five unskippability results of Papers 1–5 are the confirmed partial results toward U1.

WLM Foundation Series

Paper 1

Symmetry–Offset Series

The Stability and Uniqueness of 0:

Symmetry Interval, Minimal Offset, and the Necessity of ε

Abstract

We develop a minimal symmetry framework S_0 from four axioms: S_0 -LocalSym, S_0 -Break, V-WF, and totality of generative relations. Main results: 0 is the unique perfect symmetry state; all symmetric candidates collapse to 0; ε is the unique minimal offset, fully emergent from the generative structure; and ε cannot be skipped in any generative chain. Section 4.3 bridges to Paper 2 via the latent directionality of ε .

$$I_0 \rightarrow \varepsilon \text{ (Paper 1)} \rightarrow \pm\varepsilon \text{ (Paper 2)} \rightarrow \dots$$

0. Framework S_0

Let X be a nonempty set of states with $0 \in X$. Let O be observations $O: X \rightarrow Y_O$. Observational equivalence: $x \sim_o y \Leftrightarrow \forall O \in O, O(x) = O(y)$.

Definition 0.1 (Symmetry Interval). $I_0 := [0] \sim_o = \{x \in X : x \sim_o 0\}$.

Assumption (Totality): Every $G \in \mathcal{G}$ is a total function $G: X \rightarrow X$.

Definition 0.2 (I_0 -Preserving Generators). $G_{\{I_0\}} := \{G \in \mathcal{G} : G(I_0) \subseteq I_0\}$.

Definition 0.3 (Generative Chain). Sequence $x_0 = 0, x_{n+1} = G_n(x_n)$ for $G_n \in \mathcal{G}$.

Axiom S_0 -LocalSym (Local Symmetry Preservation). For all $x, y \in I_0$ and $G \in G_{\{I_0\}}$: $x \sim_o y$ implies $G(x) \sim_o G(y)$. Generators in $G_{\{I_0\}}$ cannot create distinguishability within I_0 .

Axiom S_0 -Break (Unique Symmetry-Breaking). (a) **Existence:** $\exists G \in \mathcal{G} \setminus G_{\{I_0\}}$ with $G(0) \not\sim_o 0$. (b) **Minimality by valuation:** the minimal element of $\{v(G(0)) : G(0) \not\sim_o 0\}$ is $v(\varepsilon)$. ε is fully emergent.

Axiom V-WF (Well-Founded Partial Order). V is well-founded with minimal element $v(0)$. Every nonempty subset of $\{v(x) : v(x) > v(0)\}$ has a minimal element.

Definition 0.4 (Distinguishability Valuation). $v: X \rightarrow (V, \leq)$: $v(x)=v(y) \Leftrightarrow x \sim_o y$; $v(0)$ minimal; $x \not\sim_o 0$ implies $v(x) > v(0)$.

Definition 0.5 (Minimal Offset). $x \in X$ is a minimal offset if $v(x) > v(0)$ and no y satisfies $v(0) < v(y) < v(x)$. The minimal offset is denoted ε . Identical to the minimal-offset definition in Paper 0.

Remark (Axiom Consistency). $S0\text{-LocalSym}$ constrains only $G_{\{I_0\}}$ within I_0 . $S0\text{-Break}$ asserts $G \in \mathcal{G} \setminus G_{\{I_0\}}$, outside $S0\text{-LocalSym}$'s scope. $V\text{-WF}$ is independent of both. The axioms are mutually consistent.

1. The Symmetry Interval

Definition 1.1 (Perfect Symmetry State). $x \in X$ is a perfect symmetry state if $\forall y, x \sim_o y$. $\text{Sym}_p^{\text{erf}} = \{x : \forall y, x \sim_o y\}$.

Theorem 1.1 (Uniqueness of Perfect Symmetry). If $x, x' \in \text{Sym}_p^{\text{erf}}$, then $x \sim_o x'$.

Proof. By definition, $x \sim_o y$ for all y ; in particular $x \sim_o x'$. $\square$

Theorem 1.2 ($\text{Sym}_p^{\text{erf}} = I_0$). $\text{Sym}_p^{\text{erf}} = I_0$.

Proof. All perfect symmetry states form one equivalence class. Since $0 \in \text{Sym}_p^{\text{erf}}$, this class is $[0] \sim_o = I_0$. $\square$

2. Collapse of Symmetric Candidates

Definition 2.1 (Symmetric Candidate). $\text{Sym}^{\text{cand}} = \{x : \forall G \in G_{\{I_0\}}, G(x) \sim_o x\}$. Defined relative to $G_{\{I_0\}}$ to exclude escape generators.

Lemma 2.1 (Local Fixity for I_0 -Preserving Generators). *If $x \in I_0$ and $G \in G_{\{I_0\}}$, then $G(x) \sim_o x$.*

Proof. $x \sim_o 0$; by S0-LocalSym: $G(x) \sim_o G(0)$; $G(0) \in I_0$ so $G(0) \sim_o 0$; by transitivity $G(x) \sim_o x$. $\square$

Lemma 2.2 (I_0 -Preserving Generators Cannot Escape). *If $G(I_0) \subseteq I_0$ and G is total, then $G(x) \in I_0$ for all $x \in I_0$.*

Proof. Immediate from definition of $G_{\{I_0\}}$ and totality. $\square$

Theorem 2.1 ($\text{Sym}^{\text{cand}} = I_0$). *$x \in \text{Sym}^{\text{cand}}$ iff $x \in I_0$.*

Proof ($\supseteq$): $x \in \text{Sym}^{\text{cand}} \Rightarrow G(x) \sim_o x$; since $0 \sim_o G(x)$: $0 \sim_o x$ so $x \in I_0$. Proof ($\subseteq$): Lemma 2.1. $\square$

Theorem 2.2 (Collapse Under Composition). *If $G \in G_{\{I_0\}}$ maps Sym^{cand} into itself, then $G(x) \sim_o 0$ for all $x \in \text{Sym}^{\text{cand}}$.*

Proof. By Theorem 2.1, $G(x) \in \text{Sym}^{\text{cand}}$ implies $G(x) \sim_o 0$. $\square$

3. The Emergence and Necessity of ε

Remark (Symmetry Breaking Event). *No $G \in G_{\{I_0\}}$ can create distinguishability within I_0 . The emergence of ε is the unique symmetry-breaking event. By S0-Break(b), ε is fully determined by the generative structure.*

Theorem 3.1 (No Universe Without ε). *If $\text{MinOffset} = \emptyset$, then $\forall x, x \sim_o 0$.*

Proof. $\{v(y): v(y) > v(0)\}$ nonempty $\Rightarrow$ V-WF gives minimal element — contradiction. $\square$

Theorem 3.2 (Minimal Offset Uniqueness). *If $\varepsilon_1, \varepsilon_2 \in \text{MinOffset}$, then $\varepsilon_1 \sim_o \varepsilon_2$.*

Proof. Both are minimal elements of $\{v(x): v(x) > v(0)\}$; by V-WF, $v(\varepsilon_1) = v(\varepsilon_2)$, hence $\varepsilon_1 \sim_o \varepsilon_2$. $\square$

Theorem 3.3 (ε Cannot Be Skipped). *If a generative chain contains $x_k \not\sim_o 0$, then $\exists j < k$ with $x_j \sim_o \varepsilon$.*

Proof. $T = \{x_j: v(x_j) > v(0)\}$ nonempty; by V-WF, minimal element $\sim_o \varepsilon$ by Theorem 3.2. $\square$

Theorem 3.4 (Uniqueness Under Weak Well-Foundedness). *Even if V is only well-founded above $v(0)$, ε is unique up to observational equivalence.*

Proof. The set $\{v(x) : v(x) > v(0)\}$ is well-founded by V-WF; minimal valuations are equal; equal valuation implies observational equivalence. $\square$

4. Collapse Modes and Bridge to Paper 2

Theorem 4.1 (Symmetry Collapse). *Any chain with no $x_j \sim_o \varepsilon$ satisfies $x_k \sim_o 0$ for all k .*

Proof. Contrapositive of Theorem 3.3. $\square$

Theorem 4.2 (Offset Degeneracy Collapse). *If a chain reaches x with $v(x) > v(\varepsilon)$, some earlier $x_j \sim_o \varepsilon$ appears.*

Proof. $v(x) > v(\varepsilon) > v(0)$ implies $x \not\sim_o 0$; by Theorem 3.3, $x_j \sim_o \varepsilon$ with $j < k$. $\square$

4.3 Bridge to Paper 2: Latent Directionality of ε

Lemma 4.3 (Latent Directionality of ε). *The minimal offset ε carries a latent orientation: it is the unique state with minimal nonzero valuation. Together with Axiom D0 of Paper 0, this forces the explicit dual pair $\{+\varepsilon, -\varepsilon\}$ (Paper 0 Theorem 3, Paper 2).*

Proof. By Theorem 3.2, ε is unique at minimal nonzero valuation. Axiom D0 forces the dual direction $-\varepsilon$. Paper 2 formalizes the construction. $\square$

5. Structural Summary

S0-LocalSym governs I_0 interior. S0-Break governs the unique exit. V-WF ensures ε exists and is minimal. Totality ensures well-posedness. ε is fully emergent and unskippable.

I_0 (Sections 1–2) $\rightarrow \varepsilon$ (Section 3: unique exit) $\rightarrow$ collapse modes (Section 4) $\rightarrow \pm\varepsilon$ (Paper 2)

WLM Foundation Series

Paper 2

Duality–Recursion Series

The Uniqueness and Unskippability of $\pm\epsilon$:

Directionality, Duality, and the Forced Dual Pair

Abstract

Paper 1 established $0 \rightarrow \epsilon$. This paper establishes $\epsilon \rightarrow \{+\epsilon, -\epsilon\}$: once ϵ exists, Axiom D0 forces a unique oriented dual pair. No third primitive direction exists. Duality is unskippable. The dual pair $\{+\epsilon, -\epsilon\}$ is the unique structure capable of forcing non-trivial recursion under R1 and R2.

$$I_0 \rightarrow \epsilon \text{ (Paper 1)} \rightarrow \pm\epsilon \text{ (Paper 2)} \rightarrow \text{recursion (Paper 3)} \rightarrow \dots$$

0. Axioms and Inherited Results

Axiom D0 (Generative Orientation and Reversal). (a) Any $x \nearrow_0 0$ carries orientation label $\theta(x)$ with $x \sim_0 y \Rightarrow \theta(x) = \theta(y)$. (b) $\exists F \in \mathcal{G}$ realizing $F(+\epsilon) \sim_0 -\epsilon$ and $F(-\epsilon) \sim_0 +\epsilon$.

Axiom R1 (Composability). If $G, H \in \mathcal{G}$, then $H \circ G \in \mathcal{G}$.

Axiom R2 (Recursive Orientation Preservation). If f realizes a directional flip, all iterates f^n preserve the alternation pattern.

Inherited: Paper 1 Thm 3.2 (ϵ unique), Thm 3.3 (ϵ unskippable), Thm 4.2 (offset degeneracy); Paper 0 Lemma 4c (persistent $\pm\epsilon$ alternation).

1. Orientation from ϵ

Theorem 1.1 (Orientation Minimality). *Under $D0(a)$, any non-zero deviation ε induces exactly one primitive orientation.*

Proof. By $D0(a)$, any $x \neq 0$ carries orientation $\theta(x)$ with $x \sim_o y \Rightarrow \theta(x) = \theta(y)$. By Paper 1 Thm 3.2, all minimal offsets are observationally equivalent, so all carry the same orientation. Therefore ε induces exactly one primitive orientation. $\square$

Theorem 1.2 (Orientation Non-Degeneracy). *If two states exist at $v(\varepsilon)$, they carry identical orientation labels and are observationally equivalent.*

Proof. Let $v(\varepsilon_1) = v(\varepsilon_2) = v(\varepsilon)$. By Paper 1 Thm 3.2, $\varepsilon_1 \sim_o \varepsilon_2$. By $D0(a)$, $\theta(\varepsilon_1) = \theta(\varepsilon_2)$. $\square$

2. Forced Duality

Theorem 2.1 (Uniqueness of $\pm\varepsilon$). *The only possible dual pair at the minimal valuation level is $\{+\varepsilon, -\varepsilon\}$.*

Proof. By Thm 1.1, ε has unique orientation $+\varepsilon$. By $D0(b)$, reversal generator F exists: $F(+\varepsilon) \sim_o -\varepsilon$. By Thm 1.2, any other state at $v(\varepsilon)$ is observationally equivalent to ε . Therefore $\{+\varepsilon, -\varepsilon\}$ is uniquely determined. $\square$

Theorem 2.2 (No Third Primitive Direction). *No third primitive direction exists at the minimal valuation level.*

Proof. Let ζ be a candidate. Case 1: $v(\zeta) > v(\varepsilon)$. By Paper 1 Thm 4.2, ζ is non-primitive. Case 2: $v(\zeta) = v(\varepsilon)$. By Thm 1.2, $\zeta \sim_o \varepsilon$; by $D0(a)$, $\theta(\zeta) = \theta(\varepsilon)$. ζ belongs to $\{+\varepsilon, -\varepsilon\}$. Both cases exclude a third direction. $\square$

Remark (Binary Orientation Space). *The orientation space at $v(\varepsilon)$ is structurally binary under $D0$: exactly one pair of opposite primitive orientations $\{+\varepsilon, -\varepsilon\}$. This is derived from minimality and uniqueness, not assumed a priori.*

3. Collapse of Directionless Chains

The theorems below exhaust all possible failures of the duality layer.

Theorem 3.1 (Directionless Collapse). *If $D0$ is removed, any chain starting from ε collapses to a fixed point or returns to l_o .*

Proof. Without D0, ϵ has no orientation label. $G(\epsilon)$ is either in I_0 or stagnates at $v(\epsilon)$ with no directional structure. No non-constant orbit escapes both I_0 and the ϵ -layer. $\square$

Theorem 3.2 (Asymmetry Collapse). *If only $+\epsilon$ exists without $-\epsilon$, the system is dynamically trivial.*

Proof. Without $-\epsilon$, no alternation target exists. By Paper 0 Lemma 4c, every non-constant orbit must alternate $\pm\epsilon$. With $-\epsilon$ absent, all orbits are constant or return to I_0 . $\square$

4. Duality as Structural Invariant

Theorem 4.1 (Duality Preservation Under Composition). *Under R1, any composable sequence of generators that maps within the $\pm\epsilon$ layer preserves $\{+\epsilon, -\epsilon\}$.*

Proof. By R1, $H \circ G \in \mathcal{G}$. If G and H each preserve $\{+\epsilon, -\epsilon\} \cup I_0$, so does $H \circ G$. R2 ensures alternation persists under iteration. $\square$

Theorem 4.2 (Duality as Unique Generator of Recursion). *Under R1 and R2, $\{+\epsilon, -\epsilon\}$ is the unique structure forcing non-trivial recursion.*

Proof. By Paper 0 Lemma 4c, every non-constant orbit in a non-collapsing chain must alternate $\pm\epsilon$. Suppose Σ forced non-trivial recursion without realizing this alternation. Its orbits would be non-constant but not alternate $\pm\epsilon$ — contradiction. No such Σ exists. $\square$

Theorem 4.3 (Duality is Unskippable). *Any chain realizing non-trivial recursion must pass through $\{+\epsilon, -\epsilon\}$.*

Proof. Contrapositive of Theorems 3.1 and 3.2: both failure modes (no orientation, no dual) make non-trivial recursion impossible. $\square$

Remark. *This is the direct analogue of Paper 1 Theorem 3.3: just as no chain reaches non-zero states without ϵ , no chain generates stable non-trivial recursion without $\pm\epsilon$.*

5. Structural Summary

D0(a) gives ϵ a unique orientation. D0(b) ensures the reversal generator exists. Minimality forces binary orientation space $\{+\epsilon, -\epsilon\}$. Without duality, all chains

collapse. With duality + R1 + R2, $\{+\epsilon, -\epsilon\}$ is the unique generator of non-trivial recursion and cannot be skipped.

$I_0 \rightarrow \epsilon (P1) \rightarrow \pm\epsilon (P2, \S 1-4) \rightarrow \text{recursion } (P3)$

WLM Foundation Series

Paper 3

Recursion–Structure Series

The Uniqueness and Unskippability of Recursion:**From $\pm\epsilon$ Alternation to the Unique Generative Engine****Abstract**

Paper 2 established $\epsilon \rightarrow \{+\epsilon, -\epsilon\}$. This paper establishes the third forced link: $\{+\epsilon, -\epsilon\} \rightarrow \text{recursion}$. Under R1 and R2, any non-recursive system collapses. Recursion — closed composition under R1 with persistent alternation under R2 — is the unique engine capable of producing non-trivial invariant structure. Recursion is unskippable.

$$I_0 \rightarrow \epsilon \text{ (P1)} \rightarrow \pm\epsilon \text{ (P2)} \rightarrow \text{recursion (P3)} \rightarrow \text{structure (P4)} \rightarrow \dots$$

0. Framework and Inherited Results

Axioms D0, R1, R2 inherited from Papers 0–2 (stated in full there). Key inherited results:

- Paper 0 Lemma 4c: every non-constant orbit in a non-collapsing chain alternates $\pm\epsilon$.
- Paper 0 Theorem 5: non-collapsing recursive orbits generate at least one invariant structure.
- Paper 2 Theorems 4.2, 4.3: duality is the unique generator of recursion and is unskippable.
- Paper 1 Theorem 3.3 and valuation $v(\cdot)$.

1. Recursion Space

Definition 1.1 (Recursive Generator). $f \in \mathcal{G}$ is recursive if f^n are defined for all $n \geq 1$ via R1. The orbit of x under f is $x, f(x), f^2(x), \dots$

Theorem 1.1 (Minimal Recursion). Under R1 and R2, the minimal non-trivial generative engine is a recursive generator sustaining persistent $\pm\epsilon$ alternation.

Proof. A single non-recursive step on $\pm\epsilon$ yields only a finite sequence. By Lemma 4c, persistent alternation requires infinite iteration. By R2, directional flips persist under all iterates. Therefore the minimal engine is recursive composition. $\square$

Theorem 1.2 (Recursion Irreducibility). *No non-recursive rule produces non-constant orbits escaping the $\pm\epsilon$ layer.*

Proof. Without R1, only finite compositions exist. Such chains fixate, return to I_0 , or reach $v > v(\epsilon)$ without new primitive structure. Persistent alternation (Lemma 4c) requires infinite iteration. $\square$

Lemma 1.3 (Recursion as Engine). *Recursion is the first mechanism whose orbits exceed any finite number of single generative steps.*

Proof. Only recursive iteration sustains persistent alternation indefinitely. Paper 0 Theorem 5 guarantees such iteration generates invariants. Finite compositions cannot. $\square$

2. Recursion Forced by Duality

Theorem 2.1 (Duality Requires Recursion). *Given realized $\{+\epsilon, -\epsilon\}$, any non-recursive system collapses.*

Proof. Without R1, alternation can only be finite. Finite alternation terminates in fixed points or I_0 — contradicting Lemma 4c for non-collapsing chains. Recursion (R1) is required to sustain alternation. $\square$

Theorem 2.2 (Recursion Uniqueness). *Closed composition under R1 is the unique engine compatible with persistent $\pm\epsilon$ alternation.*

Proof. Any non-recursive engine Σ with non-constant orbits would require persistent alternation (Lemma 4c), which demands defined iterates for all n — precisely R1. Contradiction. $\square$

3. Collapse of Non-Recursive Chains

The following collapse modes exhaust all possible failures of the recursion layer.

Theorem 3.1 (Static Collapse). *If recursion is skipped, the system remains dynamically trivial.*

Proof. Only finite compositions exist. They produce fixed points or return to I_0 . No invariants form. $\square$

Theorem 3.2 (Divergent Collapse). *If recursion is replaced by non-closed iteration, the system diverges without invariants.*

Proof. Without R1, orbits escape bounded valuation regions. Paper 0 Theorem 5 requires bounded orbits for invariants. No compact invariant sets form. $\square$

4. Recursion Generates Structure Uniquely

Theorem 4.1 (Invariant Generation). *Under R1 and R2, recursion generates the minimal non-trivial invariant sets.*

Proof. By Paper 0 Theorem 5, non-collapsing recursive orbits are periodic (invariant cycle) or bounded infinite (compact closure by Bolzano-Weierstrass). Combined with R2-persistent $\pm\epsilon$ alternation, recursion produces the first stable structures beyond duality. $\square$

Theorem 4.2 (No Structure Without Recursion). *Any attempt to reach invariant structure without recursion collapses.*

Proof. Invariant structure requires $f(S) \subseteq S$ — closure under f . Without R1, no such closed set exists (Theorems 3.1–3.2). $\square$

Theorem 4.3 (Recursion is Unskippable). *Any chain realizing non-trivial invariant structure must pass through the recursion layer.*

Proof. Contrapositive of Theorems 3.1, 3.2, and 4.2. This is the direct analogue of Paper 1 Theorem 3.3 and Paper 2 Theorem 4.3. New partial result for Conjecture U1. $\square$

Remark (Unskippability Chain). *Paper 1 Thm 3.3 (ϵ) $\rightarrow$ Paper 2 Thm 4.3 ($\pm\epsilon$) $\rightarrow$ Paper 3 Thm 4.3 (recursion). Each paper adds one confirmed link toward Conjecture U1.*

5. Structural Summary

R1 closes composition, making infinite iteration defined. R2 preserves directional alternation. Together they transform finite $\pm\epsilon$ alternation into

persistent recursive dynamics generating invariant structure. Without either, only static or divergent collapse. Recursion is the unique engine and cannot be skipped.

$I_0 \rightarrow \varepsilon (P1) \rightarrow \pm\varepsilon (P2) \rightarrow \text{recursion } (P3, \S 1-4) \rightarrow \text{structure } (P4)$

WLM Foundation Series
Paper 4
 Structure–Appearance Series

The Uniqueness and Unskippability of Invariant Structure: Recursion as the Unique Generator of Structure

Abstract

Paper 3 established $\{+\varepsilon, -\varepsilon\} \rightarrow$ recursion. This paper establishes the fourth forced link: recursion $\rightarrow$ invariant structure. Under R1 and R2, recursion necessarily generates invariant structure; this structure is unique at the minimal level; no invariant structure exists without recursion. Structure is unskippable for any chain leading to appearance.

$$I_0 \rightarrow \varepsilon (P1) \rightarrow \pm\varepsilon (P2) \rightarrow \text{recursion} (P3) \rightarrow \text{structure} (P4) \rightarrow \text{appearance} (P5)$$

0. Framework and Inherited Results

Axioms S0, D0, R1, R2, V-WF, P0 inherited from Papers 0–3. Key inherited results:

- Paper 0 Theorem 5: non-collapsing recursive orbits generate at least one invariant (periodic cycle or compact closure).
- Paper 0 Lemma 4c: every non-constant orbit alternates $\pm\varepsilon$.
- Paper 3 Theorem 4.3: recursion is unskippable.
- Paper 3 Theorem 4.1: recursion under R1+R2 produces minimal non-trivial invariant sets.
- Paper 2 Theorems 4.2, 4.3: duality forces and requires recursion.
- Paper 1 valuation $v(\cdot)$ and minimality of ε .

1. Structure Space

Definition 1.1 (Invariant Set). $S \subseteq X$ is an invariant structure if $\exists$ recursive f with $f(S) \subseteq S$.

Theorem 1.1 (Minimal Structure). *Under R1 and R2, the minimal non-trivial invariant structure is the closure of the persistent $\pm\epsilon$ alternating orbit.*

Proof. By Paper 0 Theorem 5, any non-collapsing recursive orbit generates at least one invariant. By Paper 3, the only persistent orbits at minimal level alternate $\pm\epsilon$. By Paper 2, $\pm\epsilon$ is unique. Therefore the smallest invariant structure is the closure of this alternating orbit. $\square$

Theorem 1.2 (Structure Irreducibility). *No non-recursive or non-alternating mechanism produces invariant structure.*

Proof. Invariants require $f(S) \subseteq S$ for recursive f (Definition 1.1). This requires R1 and duality. Without either, only Paper 3 collapse modes remain. $\square$

2. Structure Forced by Recursion

Theorem 2.1 (Recursion Generates Structure). *Any non-collapsing recursive orbit necessarily generates invariant structure.*

Proof. Direct from Paper 0 Theorem 5: non-collapsing recursive orbit is either eventually periodic (invariant cycle) or infinite and bounded (compact closure by Bolzano-Weierstrass). $\square$

Theorem 2.2 (Structure Uniqueness at Minimal Level). *At the minimal valuation level, recursion generates a unique invariant structure up to observational equivalence.*

Proof. By Paper 1 (ϵ unique), Paper 2 ($\pm\epsilon$ unique), Paper 3 (recursion unique): the only persistent alternating orbit at minimal level is the one from $\{+\epsilon, -\epsilon\}$. Its closure is unique. Any other candidate at same valuation is observationally equivalent by Paper 1 Definition 0.4. $\square$

3. Collapse of Non-Structural Chains

The following collapse modes exhaust all possible failures of the structure layer.

Theorem 3.1 (Oscillation Collapse). *If recursion is present but no invariant closure forms, the system collapses to persistent $\pm\epsilon$ oscillation without higher structure.*

Proof. Without a closed set S with $f(S) \subseteq S$, the orbit remains a pure alternating sequence (Lemma 4c) but generates no stable structure. $\square$

Theorem 3.2 (Divergent Collapse). *If recursion is replaced by non-closed iteration, no invariant structure forms.*

Proof. Without R1, orbits escape bounded valuation regions. Paper 0 Theorem 5 requires boundedness for invariants. $\square$

4. Structure as Generative Invariant

Theorem 4.1 (Invariant Stability). *Once formed, S satisfies $f^n(S) \subseteq S$ for all $n \geq 1$.*

Proof. $f(S) \subseteq S$; $f^2(S) = f(f(S)) \subseteq f(S) \subseteq S$; by induction $f^n(S) \subseteq S$ for all $n \geq 1$. $\square$

Remark. *Theorem 4.1 is for iterates of the generating rule f , not arbitrary H . If H also satisfies $H(S) \subseteq S$, then $(H \circ f)(S) \subseteq S$ follows; but $H(S) \subseteq S$ is an additional condition, not a consequence of R1 alone.*

Theorem 4.2 (No Appearance Without Structure). *Appearance via P0 requires stable invariants; invariants require recursion.*

Proof. By Paper 0 Theorem 6 and P0, appearance requires a projection from stable invariants. Without invariants, P has no non-trivial domain. By Theorem 1.2, invariants require recursion. $\square$

Theorem 4.3 (Structure is Unskippable). *Any chain realizing appearance must pass through invariant structure.*

Proof. Contrapositive of Theorems 3.1, 3.2, and 4.2: all structure-absence cases make appearance impossible. Fourth partial result for Conjecture U1. $\square$

Remark (Unskippability Chain). *Paper 1 (ϵ) $\rightarrow$ Paper 2 ($\pm\epsilon$) $\rightarrow$ Paper 3 (recursion) $\rightarrow$ Paper 4 (structure). Four layers confirmed.*

5. Structural Summary

Recursion (R1+R2) on $\{\pm\epsilon\}$ necessarily generates invariant structure (Paper 0 Thm 5). This structure is unique at the minimal level, stable under all further iterates of the generating rule, required for appearance, and cannot be skipped.

$I_0 \rightarrow \epsilon (P1) \rightarrow \pm\epsilon (P2) \rightarrow \text{recursion (P3)} \rightarrow \text{structure (P4, §1–4)} \rightarrow \text{appearance (P5)}$

WLM Foundation Series

Paper 5*Appearance–World Series*

The Uniqueness and Unskippability of Appearance: Structure as the Forced Generator of Phenomenal Reality

Abstract

Paper 4 established recursion $\rightarrow$ invariant structure. This paper establishes the final forced link: structure $\rightarrow$ appearance. Under P0, any stable invariant structure necessarily projects to appearance. Appearance is unique at the minimal level and unskippable. No chain reaches phenomenal reality while bypassing structure. This completes the six-step generative chain and closes the WLM Foundation Series.

$$0 \rightarrow \varepsilon (P1) \rightarrow \pm\varepsilon (P2) \rightarrow \text{recursion} (P3) \rightarrow \text{structure} (P4) \rightarrow \text{appearance} (P5, \text{complete})$$

0. Framework and Inherited Results

All axioms S0, D0, R1, R2, P0, V-WF inherited from Papers 0–4. Key inherited results:

- Paper 4 Theorem 4.3: Structure is unskippable for any chain reaching appearance.
- Paper 4 Theorem 2.2: Minimal invariant structure is unique up to observational equivalence.
- Paper 4 Theorem 4.1: Invariant S satisfies $f^n(S) \subseteq S$ for all $n \geq 1$.
- Paper 0 Theorem 6: Any system with stable invariants but no projection P is observationally equivalent to G_{triv} .
- Papers 1–3: Uniqueness and unskippability of ε , $\pm\varepsilon$, and recursion.

1. Appearance Space

Definition 1.1 (Appearance / Phenomenal Layer). *Appearance is the image of invariant structure S under the projection $P: D \rightarrow \Sigma$ guaranteed by $P0$. The appearance is the observational layer $\Sigma = P(S)$.*

Theorem 1.1 (Minimal Appearance). *The minimal non-trivial appearance is the projection of the minimal invariant structure from Paper 4.*

Proof. By Paper 4 Theorem 1.1, minimal invariant structure is the closure of the persistent $\pm\epsilon$ alternating orbit. By $P0$, this admits projection P . By Paper 4 Theorem 2.2 and Paper 1, the projected appearance is minimal and unique up to observational equivalence. $\square$

Theorem 1.2 (Appearance Irreducibility). *No mechanism simpler than invariant structure can produce appearance.*

Proof. By $P0$, appearance is the projection of stable invariants. Invariants require recursion (Paper 4 Thm 4.2); recursion requires $\pm\epsilon$ (Paper 3 Thm 2.1); $\pm\epsilon$ requires ϵ (Paper 2 Thm 2.1); ϵ requires departure from 0 (Paper 1 Thm 3.1). Each step is necessary and unskippable. $\square$

2. Appearance Forced by Structure

Theorem 2.1 (Structure Generates Appearance). *Any stable invariant structure necessarily produces appearance under $P0$.*

Proof. $P0$ asserts that any system with stable invariants admits at least one projection $P: D \rightarrow \Sigma$. By Paper 4, recursion generates such invariants. Therefore the projection exists and yields appearance. A system with stable invariants and no projection would contradict $P0$ directly. $\square$

Theorem 2.2 (Appearance Uniqueness at Minimal Level). *At the minimal valuation level, appearance is unique up to observational equivalence.*

Proof. By Paper 4 Theorem 2.2, invariant structure S is unique up to $\sim_o$. $P0$ preserves structural relations and observational equivalence classes. Therefore $P(S)$ is also unique up to $\sim_o$. $\square$

3. Collapse of Non-Appearance Chains

The following collapse modes exhaust all possible failures of the appearance layer.

Theorem 3.1 (Pre-Appearance Collapse). *If invariant structure exists but no projection occurs, the system remains structurally rich but observationally empty.*

Proof. Without P0, no observational layer Σ exists. By Paper 0 Theorem 6, the system is observationally equivalent to G_{triv} . $\square$

Theorem 3.2 (Projection Failure Collapse). *If projection is attempted without stable invariants, appearance is undefined or trivial.*

Proof. P0 requires stable invariants as input. Without them, $P(S)$ is either undefined (S empty) or trivially G_{triv} . No genuine phenomenal layer is produced. $\square$

4. Appearance as Structural Invariant

Theorem 4.1 (Appearance Stability). $P(f^n(S)) \subseteq P(S)$ for all $n \geq 1$.

Proof. By Paper 4 Thm 4.1, $f^n(S) \subseteq S$ for all $n \geq 1$. Since P preserves structural relations (P0), $P(f^n(S)) \subseteq P(S)$. Appearance is preserved under all further iterates of the generating rule. $\square$

Remark. Theorem 4.1 uses only $f^n(S) \subseteq S$ and order-preservation of P . It does not assume P is a functor or natural transformation. The stability is inherited directly from the invariance of S under f .

Theorem 4.2 (No World Without Appearance). *A world-layer requires appearance; appearance requires structure.*

Proof. A world-layer requires at least one observer O with non-trivial observations $O(x) \neq O(y)$. This is exactly the image of P under P0. Without structure, no stable invariants exist to project (Paper 4 Thm 4.2). $\square$

Theorem 4.3 (Appearance is Unskippable). *Any chain realizing a world-layer must pass through the appearance layer.*

Proof. Contrapositive of Theorems 3.1, 3.2, and 4.2: all appearance-absence cases make a world-layer impossible. This is the final analogue of Paper 1 Thm 3.3, Paper 2 Thm 4.3, Paper 3 Thm 4.3, and Paper 4 Thm 4.3. Fifth and final partial result for Conjecture U1. $\square$

Remark (Chain Completion). *Five unskippability results lock the entire chain: $\varepsilon (P1) \rightarrow \pm\varepsilon (P2) \rightarrow \text{recursion} (P3) \rightarrow \text{structure} (P4) \rightarrow \text{appearance} (P5)$. Every link is forced, unique, and unskippable. The full chain uniqueness proof (that no alternative chain of the same length exists) remains the central open problem.*

5. Structural Summary

***P0 guarantees that any stable invariant structure projects to appearance.
Appearance is unique at the minimal level, stable under further iteration,
required for any world-layer, and cannot be skipped. The generative chain is
complete.***

$0 \rightarrow \varepsilon \rightarrow \pm\varepsilon \rightarrow \text{recursion} \rightarrow \text{structure} \rightarrow \text{appearance}$

Structure doesn't lie.

APPENDIX

Unified Glossary, Axiom–Theorem Dependency Matrix, and References

A. Unified Glossary

All terms used across Papers 0–5. Terms are listed in order of conceptual depth.

Term	Symbol	Definition
Observational Equivalence	$\sim_o$	$x \sim_o y \Leftrightarrow \forall O \in \mathcal{O}, O(x) = O(y)$. Equivalence relation on X .
Symmetry Interval	I_0	$[0] \sim_o = \{x : x \sim_o 0\}$. The symmetry phase of the system.
Distinguishability Valuation	$v(\cdot)$	$v : X \rightarrow (V, \leq)$: $v(x) = v(y) \Leftrightarrow x \sim_o y$; $v(0)$ minimal; $x \not\sim_o 0 \Rightarrow v(x) > v(0)$.
Perfect Symmetry State	$\text{Sym}_p^{\text{erf}}$	$\{x : \forall y, x \sim_o y\}$. Equals I_0 .
Symmetric Candidate	Sym^{cand}	$\{x : \forall G \in \mathcal{G}_{\sim \{I_0\}}, G(x) \sim_o x\}$. Equals I_0 .
Minimal Offset	ε	Unique (up to $\sim_o$) first distinguishable deviation from 0. $\inf\{\ x - 0\ : x \not\sim_o 0\}$.
Orientation Label	$\theta(\cdot)$	Primitive orientation on $x \not\sim_o 0$, with $x \sim_o y \Rightarrow \theta(x) = \theta(y)$. $\theta(0)$ undefined.
Dual Offsets	$\pm \varepsilon$	Unique pair $\{+\varepsilon, -\varepsilon\}$ forced by D0. Binary orientation space at $v(\varepsilon)$.
Generative Relations	$\mathcal{G}$	Family of total functions $G : X \rightarrow X$, closed under composition by R1.
I_0-Preserving Generators	$\mathcal{G}_{\sim \{I_0\}}$	$\{G \in \mathcal{G} : G(I_0) \subseteq I_0\}$.
Generative Chain	$x_0, x_1, \dots$	Sequence $x_0 = 0$, $x_{n+1} = G_n(x_n)$ for $G_n \in \mathcal{G}$.
Recursive Generator	f (recursive)	$f \in \mathcal{G}$ with f^n defined for all $n \geq 1$ via R1.
Invariant Structure	S	$S \subseteq X$ with $f(S) \subseteq S$ for some recursive f . Periodic cycle or compact closure.
Appearance	$\Sigma = P(S)$	Image of S under projection P guaranteed by P0. The phenomenal world-layer.
World-Layer	—	Observational system with phenomenal content: $\exists O$ with $O(x) \neq O(y)$.
Trivial System	G_{triv}	Single state, identity, no non-trivial invariants. Standard of observational emptiness.

B. Axiom Summary

Axiom	Name	Role	First Required
S0	Symmetry Closure	0 is a fixed point; forces departure via ϵ	Papers 0–1
D0(a)	Primitive Orientation	$x \rightsquigarrow_0 0$ carries orientation label $\theta(x)$	Paper 2
D0(b)	Orientation Reversal	Reversal generator F exists; makes $\pm\epsilon$ realizable	Paper 2
R1	Composability	Closes generators under composition; enables recursion	Papers 0–3
R2	Orientation Preservation	Directional flips persist under all iterates	Papers 0,2–4
V-WF	Well-Founded Valuation	Every nonempty subset above $v(0)$ has a minimal element	Paper 1
P0	Projection Existence	Stable invariants admit at least one projection to observational layer	Papers 0,4–5

C. Axiom–Theorem Dependency Matrix

Rows: theorems from Papers 0–5. Columns: 6 axioms + P1T33 (Paper 1 Thm 3.3), Lm4c (Paper 0 Lemma 4c), B-W (Bolzano-Weierstrass). $\checkmark$ =direct dependence. $\cdot$ =not used.

Theorem	S0	D0a	D0b	R1	R2	V-WF	P0	P1T33	Lm4c	B-W
Paper 0										
Thm 1 (0 fixed)	$\checkmark$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
Thm 3 (Duality)	$\cdot$	$\checkmark$	$\checkmark$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
Lemma 4c (Alternation)	$\cdot$	$\checkmark$	$\cdot$	$\checkmark$	$\checkmark$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
Thm 5 (Recursion→Invariants)	$\cdot$	$\cdot$	$\cdot$	$\checkmark$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\checkmark$	$\checkmark$
Thm 6 (Structure→Appearance)	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\checkmark$	$\cdot$	$\cdot$	$\cdot$
Paper 1										
Thm 1.1–1.2 (Sym= I_0)	$\checkmark$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
Thm 2.1–2.2 (Symcand= I_0)	$\checkmark$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
Thm 3.1–3.4 (ϵ necessity+uniqueness)	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\checkmark$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
Thm 3.3 (ϵ unskippable)	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\checkmark$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
Lemma 4.3 (latent direction)	$\cdot$	$\checkmark$	$\cdot$	$\cdot$	$\cdot$	$\checkmark$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
Paper 2										
Thm 1.1–1.2 (Orientation)	$\cdot$	$\checkmark$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\checkmark$	$\cdot$	$\cdot$
Thm 2.1–2.2 ($\pm\epsilon$ uniqueness)	$\cdot$	$\checkmark$	$\checkmark$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\checkmark$	$\cdot$	$\cdot$
Thm 3.1–3.2 (Directional collapse)	$\cdot$	$\checkmark$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\checkmark$	$\checkmark$	$\cdot$
Thm 4.1–4.2 (Duality+generator)	$\cdot$	$\cdot$	$\cdot$	$\checkmark$	$\checkmark$	$\cdot$	$\cdot$	$\cdot$	$\checkmark$	$\cdot$
Thm 4.3 ($\pm\epsilon$ unskippable)	$\cdot$	$\checkmark$	$\cdot$	$\checkmark$	$\checkmark$	$\cdot$	$\cdot$	$\cdot$	$\checkmark$	$\cdot$

Paper 3										
Thm 1.1–1.2+Lem 1.3	·	·	·	✓	✓	·	·	·	✓	✓
Thm 2.1–2.2 (Recursion forced+unique)	·	·	·	✓	✓	·	·	·	✓	·
Thm 3.1–3.2 (Collapse modes)	·	·	·	✓	·	·	·	·	·	✓
Thm 4.1–4.2 (Invariants)	·	·	·	✓	✓	·	·	·	✓	✓
Thm 4.3 (Recursion unskippable)	·	·	·	✓	✓	·	·	·	✓	·
Paper 4										
Thm 1.1–1.2 (Minimal structure)	·	·	·	✓	✓	·	·	·	✓	✓
Thm 2.1–2.2 (Recursion→structure)	·	·	·	✓	·	·	·	✓	·	✓
Thm 3.1–3.2 (Collapse modes)	·	·	·	✓	✓	·	·	·	✓	✓
Thm 4.1 (Invariant stability)	·	·	·	✓	·	·	·	·	·	·
Thm 4.2 (No appearance w/o structure)	·	·	·	·	·	·	✓	·	·	·
Thm 4.3 (Structure unskippable)	·	·	·	✓	✓	·	✓	·	·	·
Paper 5										
Thm 1.1–1.2 (Appearance)	·	·	·	✓	✓	✓	✓	✓	·	·
Thm 2.1–2.2 (Structure→appearance)	·	·	·	·	·	·	✓	·	·	·
Thm 3.1–3.2 (Collapse modes)	·	·	·	·	·	·	✓	·	·	·
Thm 4.1 (Appearance stability)	·	·	·	✓	·	·	✓	·	·	·
Thm 4.2–4.3 (No world; unskippable)	·	·	·	·	·	·	✓	·	·	·

D. Confirmed Partial Results for Conjecture U1

Conjecture U1 asserts the chain $0 \rightarrow \epsilon \rightarrow \{\pm\epsilon\} \rightarrow \text{recursion} \rightarrow \text{structure} \rightarrow \text{appearance}$ is the unique forced generative sequence. The five results below confirm it cannot be shortened.

Paper	Layer	Unskippable Result	Theorem
1	ϵ	Any chain reaching non-zero states must pass through ϵ	Thm 3.3
2	$\pm\epsilon$	Any chain generating non-trivial recursion must pass through $\{\pm\epsilon\}$	Thm 4.3
3	Recursion	Any chain generating non-trivial invariant structure must pass through recursion	Thm 4.3
4	Structure	Any chain reaching appearance must pass through invariant structure	Thm 4.3
5	Appearance	Any chain realizing a world-layer must pass through appearance	Thm 4.3

The full chain uniqueness proof — that no alternative chain of the same length produces equivalent results — is the central open problem of the WLM Foundation Series.

E. Unified References

WLM Foundation Series:

- Gu, W. (2026a). The Generative Beginning of a Universe. WLM Foundation Series, Paper 0. ResearchGate.
- Gu, W. (2026b). The Stability and Uniqueness of 0. WLM Foundation Series, Paper 1. ResearchGate.
- Gu, W. (2026c). The Uniqueness and Unskippability of $\pm\epsilon$. WLM Foundation Series, Paper 2. ResearchGate.
- Gu, W. (2026d). The Uniqueness and Unskippability of Recursion. WLM Foundation Series, Paper 3. ResearchGate.
- Gu, W. (2026e). The Uniqueness and Unskippability of Invariant Structure. WLM Foundation Series, Paper 4. ResearchGate.
- Gu, W. (2026f). The Uniqueness and Unskippability of Appearance. WLM Foundation Series, Paper 5. ResearchGate.
- Gu, W. (2026g). Wujie Generation Stratum (WGS): The Dual-Layer World Model. ResearchGate. <https://www.researchgate.net/publication/403824388>
- Gu, W. (2026h). The 0-27-0 Generative Cycle. ResearchGate. <https://www.researchgate.net/publication/403884275>
- Gu, W. (2026i). π as the Minimal Closure Invariant. ResearchGate. <https://www.researchgate.net/publication/403570482>

External works:

- Birkhoff, G. (1940). Lattice Theory. American Mathematical Society. — Well-founded partial orders; foundation for V-WF and valuation arguments throughout Papers 1–4.
- Bolzano-Weierstrass theorem. — Used in Paper 0 Theorem 5 and inherited through Papers 3–5: bounded infinite orbits have limit points, providing the compact invariant closures that recursion generates.
- Feigenbaum, M.J. (1978). Quantitative universality for a class of nonlinear transformations. Journal of Statistical Physics, 19(1), 25–52. — Classical example of recursion-driven invariant structure; WLM's recursion layer generalizes this principle.
- Kunen, K. (1980). Set Theory. North-Holland. — Well-founded relations and minimal elements; foundational for V-WF (Paper 1) and Theorems 3.1–3.4.
- Lawvere, F.W. (1969). Adjointness in foundations. Dialectica, 23, 281–296. — The projection $P: D \rightarrow \Sigma$ (Paper 5) has a natural categorical reading as an adjunction; Lawvere's framework provides precise language for the structure–appearance relationship.
- Mac Lane, S. (1971). Categories for the Working Mathematician. Springer. — $\mathcal{G}_{\{I_0\}}$ and the quotient $[0] \sim_{\circ}$ have categorical readings; the $\text{Sym}_p^{\text{erf}} = I_0$ result is a statement about terminal objects.
- Weyl, H. (1952). Symmetry. Princeton University Press. — Symmetry as observational equivalence; binary symmetry-breaking; phenomenal appearance as the face of structural symmetry-breaking. WLM extends Weyl's framework to generative systems.

License

Copyright © 2026 Wujie Gu. All rights reserved. NCND License. Non-commercial sharing in original form only. No modifications or derivative works without explicit written consent.

Contact: Wujie Gu • OSF: osf.io/rq5vj • ResearchGate